ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE G · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice g.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers and wording. Circuits may be drawn in different but correct ways: mark each required element and correct connection.

QUESTION 1

1.1 Astable (free-running) ✓ – flashing lights, clock pulses ✓; monostable (one-shot) ✓ – timers, delay circuits, switch debouncing ✓; bistable (flip-flop) ✓ – memory / latch, toggle on-off control ✓ (6)

1.2 The SCR is placed in series with the motor and the supply, anode to + ✓; a small pulse on the gate switches it on and the motor runs ✓; it stays on even when the gate pulse stops ✓; it switches off only when the anode current falls below the holding current (e.g. the supply is interrupted or reversed / a stop switch breaks the circuit) ✓. (4)

1.3 An inductor (coil) stores energy in its magnetic field and opposes any change in current ✓✓. Application ✓: filter / choke in power supplies, relay and solenoid coils, tuned circuits, motor windings. (3)

1.4 Inputs (sensors, switches) ✓ → input interface / ADC ✓ → microcontroller (CPU, memory, I/O) ✓ → output interface / drivers ✓ → outputs (motors, lamps, valves) ✓ (a power supply block may also be shown) (5)

1.5 The maximum value (height) of a waveform ✓, measured from the zero (centre) line to the peak ✓. (2)

[20]

QUESTION 2

2.1 Any FIVE ✓✓✓✓✓: clean and quiet; high efficiency; accurate speed and position control; easy to control from PLCs and computers; quick start, stop and reversal; low maintenance; available in many sizes; no exhaust gases; regenerative braking possible. (5)

2.2 The controller sends the wanted position (set point) ✓; a feedback device (potentiometer / encoder) on the shaft measures the actual position ✓; the controller compares the two and drives the motor to reduce the difference (error) ✓; when the error is zero the motor stops and holds the position, correcting any disturbance ✓. (4)

2.3 Op-amp triangle with − and + inputs ✓; input resistor R_{in} to the inverting (−) input ✓; feedback resistor R_f from output to the − input ✓; + input to 0 V / ground ✓; input and output labelled ✓. Gain = $-R_f \div R_{in}$, set by choosing the two resistors ✓ (6)

2.4 $\text{Gain} = -R_f \div R_{in} = -100 \div 10 = -10$ ✓; $V_{out} = -10 \times 0{,}5$ ✓ $= -5$ V ✓ (3)

2.5 The process of writing, testing and correcting a set of instructions (a program) ✓ that tells a computer or controller what to do ✓. (2)

[20]

QUESTION 3

3.1 It keeps back-pressure on the cylinder outlet ✓ so that a hanging or vertical load cannot fall or run away by itself ✓; the load moves down only when pressure is applied to drive it, giving smooth, controlled lowering ✓. (3)

3.2 Any FIVE ✓✓✓✓✓: force needed (load); stroke length; bore (piston) diameter; rod diameter / buckling; working pressure; speed required; single- or double-acting; mounting method; type of seals / operating temperature; cushioning. (5)

3.3 Correct symbol (vertical capsule / oval with a line across) ✓✓. It stores oil under pressure ✓ and releases it when needed – to supply extra flow, keep pressure steady, absorb shocks or provide emergency power ✓. (4)

3.4 Tank, pump and motor ✓; pressure relief valve to tank ✓; pressure gauge on the pressure line ✓; 4/3 lever valve with springs on both sides ✓ and a tandem centre (P to T, A and B blocked) ✓; double-acting cylinder with A and B correctly connected ✓; filter in the return line to tank ✓; correct ISO symbols and neat ✓ (8)

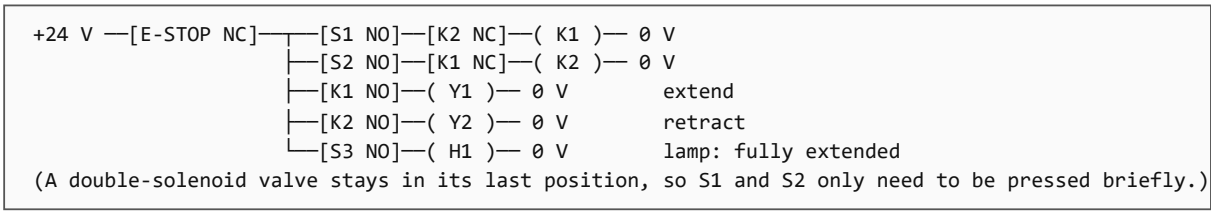
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QUESTION 4

4.1 A pressure relief valve is normally closed ✓ and opens to send oil back to tank when the system pressure reaches its setting, protecting the whole system ✓. A pressure-reducing valve is normally open ✓ and gives a lower, constant pressure to one branch of the circuit (e.g. a clamp) while the rest works at full pressure ✓. (4)

4.2 An interlock is a contact of one relay placed in the circuit of another ✓ so that one action is blocked while the other is active ✓; it prevents Y1 and Y2 from being energised together, which would make the valve work against itself and could burn out a coil ✓. (3)

4.3 Supply lines ✓; E-stop (NC) in the main supply ✓; S1 (NO) ✓ in series with NC contact K2 ✓ driving K1 ✓; S2 (NO) ✓ in series with NC contact K1 ✓ driving K2 ✓; K1 contact switching Y1 and K2 contact switching Y2 ✓; S3 (NO) switching lamp H1 ✓; correct symbols and neat ✓, e.g. (11)



4.4 Any TWO ✓✓: limit switches (roller lever); inductive proximity sensors; reed switches / magnetic cylinder sensors; photoelectric sensors; linear position transducers. (2)

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QUESTION 5

5.1 It disconnects the electrical supply completely ✓ so that the machine cannot be started while someone is working on it ✓; it can be locked off and tagged so that only the person working can switch it on again ✓. (3)

5.2 Air enters through a low oil level (pump sucking air), loose or leaking suction fittings, worn shaft seals, or oil returning above the oil level and foaming ✓✓. Effects ✓✓: spongy, jerky movement of actuators; noise (cavitation) and pump damage; oil foaming and overheating; poor control of speed and position. (4)

5.3 Any THREE ✓✓✓: cavitation from a blocked suction filter or suction line; air in the oil (aeration); low oil level; oil too thick (cold or wrong grade); worn pump bearings or parts; pump driven too fast; loose mounting / misalignment of the coupling. (3)

5.4 Any SEVEN steps in order ✓✓✓✓✓✓✓: switch off and lock out the motor isolator; release all pressure in the system; place a drip tray and mark the hoses; disconnect the suction and pressure lines and plug all openings; remove the coupling guard and unbolt the pump; fit the new pump with the correct rotation direction and align the coupling; reconnect the lines and fill the pump with clean oil (prime it); top up the tank, start at low pressure, bleed air and check for leaks and noise; set the relief valve and test the system; remove the lock-out and record the work. (7)

5.5 They show the state of the machine or circuit at a glance ✓ (running, stopped, faulted, power on) and help with fault-finding ✓. Example ✓: green lamp = motor running, red lamp = fault / stopped, white lamp = control power on. (3)

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TOTAL: 100

END OF MARKING GUIDELINE